

Participant's Manual

Displays, Indicators and Controls E90



The information contained in this Participant's Manual is intended solely for the participants of this seminar run by BMW Aftersales Training.

Refer to the latest relevant "BMW Service" information for any changes/supplements to the Technical Data.

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Participant's Manual

Displays, Indicators and Controls

E90

Concise information at a glance

Controller also in the BMW 3 Series

More discrete appearance of the interior through less switches



Information on this Participant's Manual

Symbols used

The following symbols are used in this Participant's Manual to facilitate better comprehension and to draw attention to important information.

 contains information for better understanding of the described systems and their functions.

◀ identifies the end of an item of information.

Current content of Participant's Manual

In view of the constant further developments in the design and equipment of BMW vehicles deviations may arise between this Participant's Manual and the vehicles made available as part of the training course.

The background material refers exclusively to left-hand drive vehicles. The controls are in part arranged differently in right-hand drive vehicles than shown on the graphics in the Participant's Manual.

Additional information sources

You will find further information on the individual vehicle topic in the BMW diagnosis and repair systems as well as on the Internet under www.bmw.com.

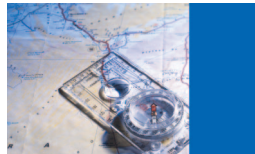
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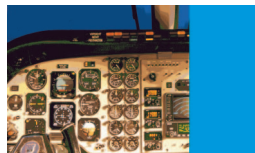
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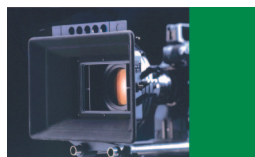
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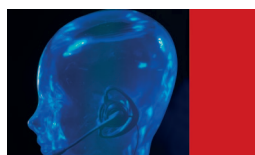


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Objectives

Displays, Indicators and Controls

Reference material for practical applications to accompany you throughout the training course

This Participant's Manual provides information on the compact operational control concept as well as the ergonomically designed and well-positioned display, indicator and control elements in the new BMW 3 Series.

This manual is designed to provide essential information throughout the training course and complements the seminar material used in the BMW Aftersales Training course. It can be used both as a basis for self-study as well as a reference document.

While preparing for the technical training course and in connection with the practical exercises, this information is designed to enable the participant to perform service work in the area of the display, indicator and control elements on the new BMW 3 Series.

Previous technical and practical knowledge of the BMW 1 Series, 5 Series and 7 Series will facilitate better understanding of the corresponding systems and their functions.



It is important that you work through the training and information program (SIP) on this subject. Basic knowledge ensures competence in theory and practical applications.



Introduction

Displays, Indicators and Controls

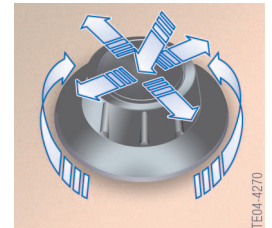
An inspired concept

Clear differentiation and optimum structuring of the cockpit are the prerequisite for the more discrete and subtle appearance of the interior. Fewer switches simplify logical operation in the new BMW 3 Series.

The display, indicator and control elements are organized in a hierarchical arrangement corresponding to their function - "Most important at the top - less important at the bottom"

The Participant's Manual is subdivided into the following sections:

- Instrument cluster
- Central Information Display (CID)
- ConnectedService
- Personal profile.



Depending on the equipment configuration, it is possible to operate all comfort functions and several vehicle functions via the controller in the BMW 3 Series.

Display, indicator and control concept



1 - BMW 3 Series display, indicator and control concept

Index	Description	Index	Description
1	Steering column stalk/steering wheel	3	Central Information Display (CID)
2	Instrument cluster	4	Controller



A classic two-dial design draws attention to both instruments. Each finished unobtrusively with two trim rings in pearl-finish chrome.

Instrument Cluster

The new BMW 3 Series is equipped with an instrument cluster featuring analogue instruments for:

- Speed
- Engine speed
- Economy control and
- Fuel level.

Two large pointer instruments show the road speed and engine speed.

Two smaller pointer instruments display fuel level and current fuel consumption.

The scales in the instrument cluster are specific to the country, vehicle and engine.

All of the indicator lamps are located in the centre at the top between the two large instrument dials.

The LC display is located in the centre between the two large pointer instruments.



2 - Basic instrument cluster, diesel

Central Information Display (CID)

The central information display (CID) is the graphic display unit for the user interface of all convenience functions and several vehicle functions.

The main menu consists of four menu items:

- Communication
- Navigation
- Entertainment
- Climate control.

All individual user settings are combined under the additional "Settings" menu.

The button for selecting the main menu is located behind the controller in the centre console.

As on the mid-range and luxury BMW models, the system is operated by means of the central control element, the controller.



The new BMW 3 Series offers the same convenient operating and control functions as the mid-range and luxury BMW models.



3 - Central Information Display (CID)

The design of the central information display is identical to that of the CID installed in the BMW 5 Series.

In the BMW 3 Series, the same software is used in the central information display as in the BMW 5 Series and in the BMW 1 Series. All functions are identical and are described in the BMW 3 Series Owner's Handbook.



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The KeyReader is an important ConnectedService module. Service reception is able to read the data stored in the vehicle key with the KeyReader.

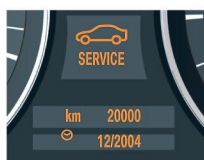
ConnectedService

ConnectedService has been part of the service concept of the BMW Group since the introduction of the BMW 7 Series.

ConnectedService is made up of several modules; some of these are already in operation while others have been further developed or introduced for the first time with the launch of the BMW 3 Series.

ConnectedService is an indication of how communications and networking between the car and Service are increasing.

TeleService1 can even make an automatic ServiceCall which informs the customer's home dealer that a service is due.



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4 - ConnectedService modules

Personal profile

The "Personal profile" system allows the driver to set several functions of the BMW 3 Series to suit his/her personal requirements.

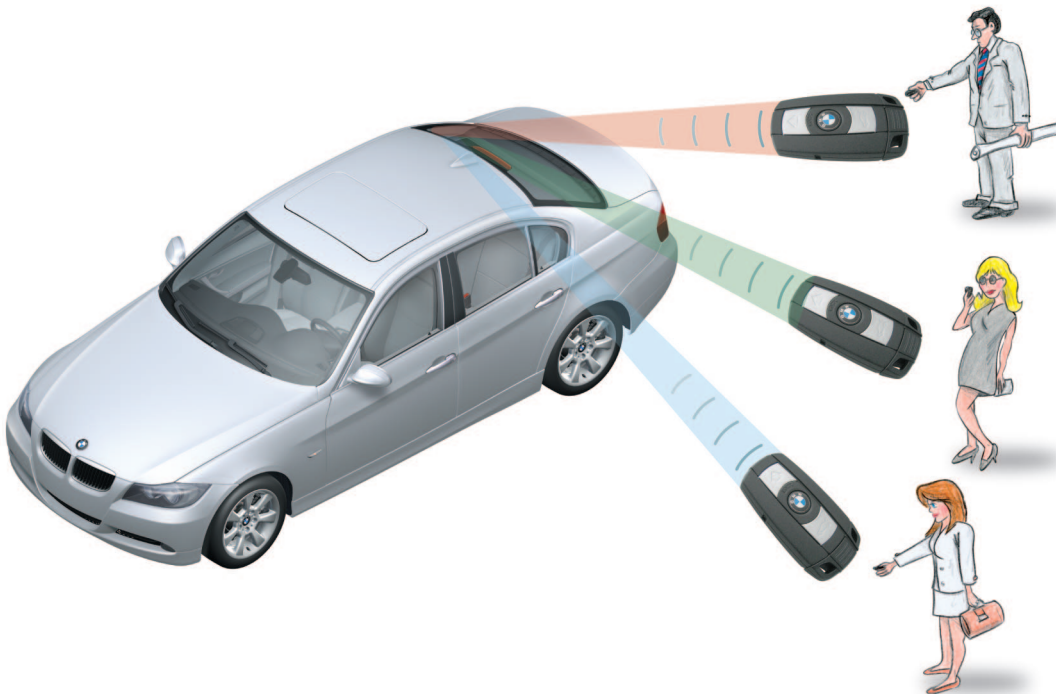
Personal profile stores the data entered by the driver such as automatic setting of the outside mirrors or speed-dependent volume in the corresponding control units.

As soon as the vehicle is unlocked using the remote control, the system recognizes the corresponding settings belonging to the remote control.

Up to three different basic settings can be adapted for three different persons. The precondition is that each of the three persons has his/her own remote control.



The BMW 3 Series caters for the driver's personal requirements.



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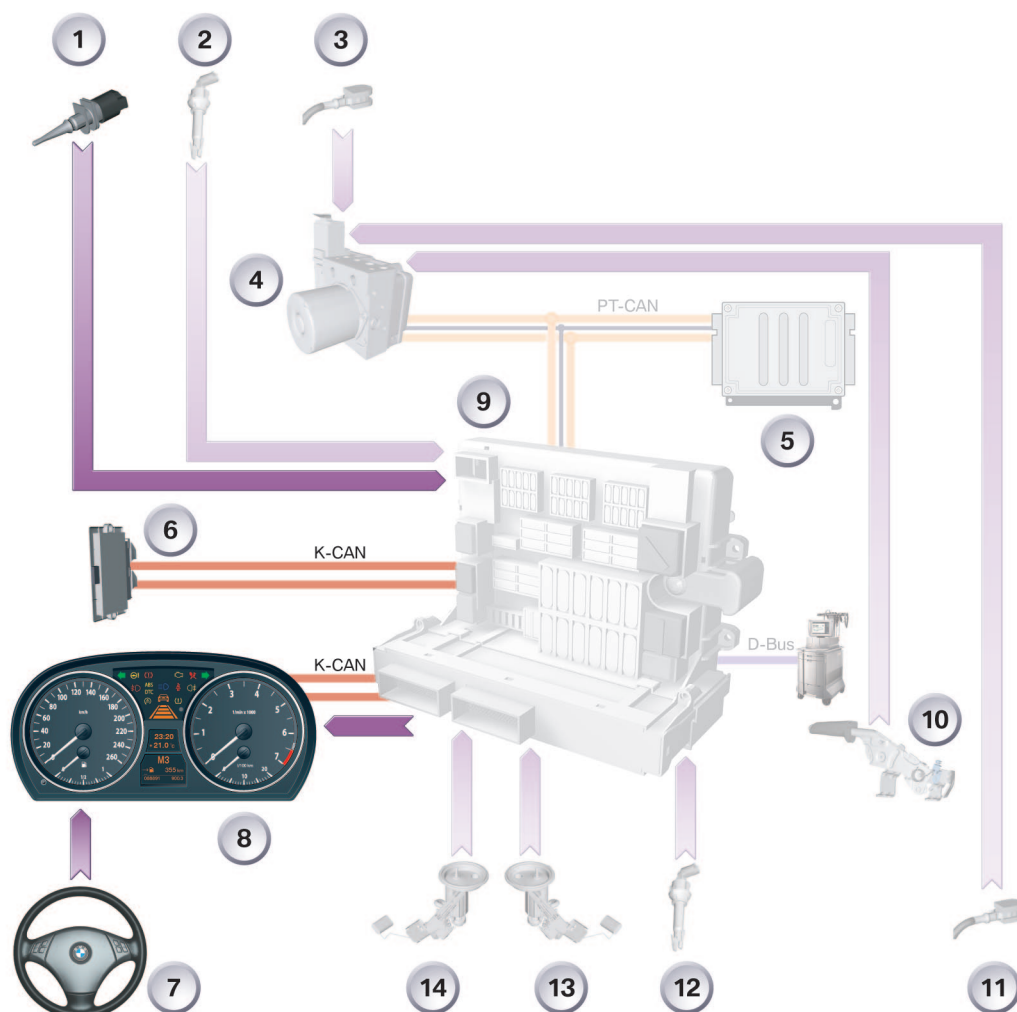


System overview

Displays, Indicators and Controls

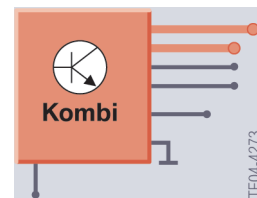
Instrument cluster

Input/output



1 - Input/output

The input/output key is identical to that of the system circuit diagram and is provided after the system circuit diagram.



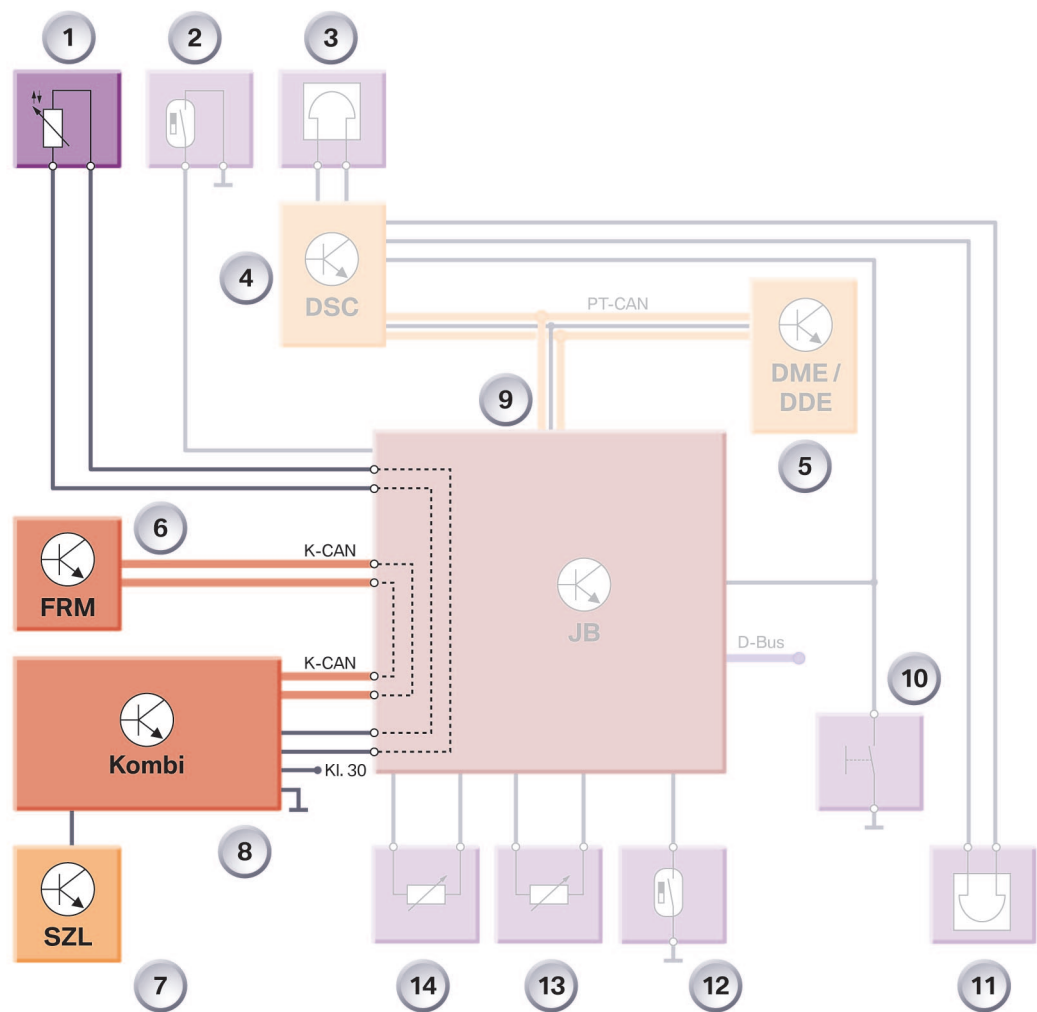
Unlike the previous model variants, the instrument cluster in the BMW 3 Series has no gateway function.

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System circuit diagram

As can be clearly seen in the system circuit diagram, the instrument cluster on the BMW 3 Series has no gateway function.

Only the outside temperature, the information from the steering column switch cluster and footwell module are shown directly in the display.



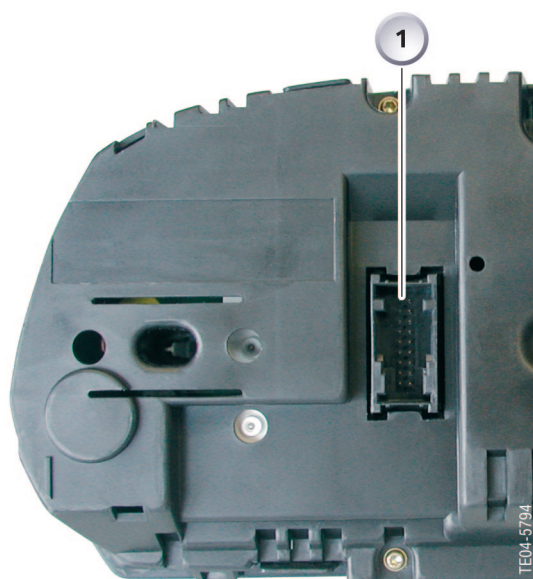
2 - System circuit diagram

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Index	Description	Index	Description
1	Outside temperature sensor	11	Brake pad wear sensor, rear right
2	Coolant level switch	12	Washer fluid level sensor
3	Brake pad wear sensor, front left	13	Fuel level sensor, right
4	Dynamic Stability Control (DSC)	14	Fuel level sensor, left
5	Digital motor electronics DME or DDE	PT-CAN	Powertrain Controller Area Network
6	Footwell module FRM	D-Bus	Diagnostics bus
7	Steering column switch cluster	K-CAN	Body Controller Area Network
8	Instrument cluster	Kl. 30	Terminal 30 (power supply)
9	Junction box JB	Kl. 31	Terminal 31 (ground)
10	Contact switch, handbrake		

Connector assignments of the instrument cluster

The 18-pin plug connection (1) is located at the rear of the instrument cluster.

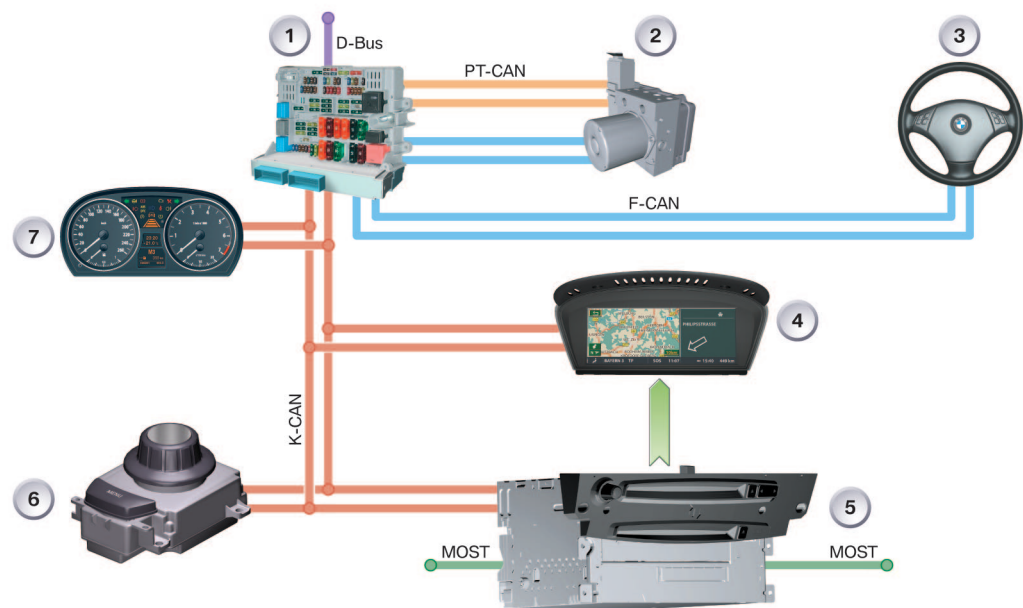


3 - 18-pin plug connection

Central Information Display (CID)

Input/output

The following illustration shows all control units and control elements that are connected directly to the central information display CID.

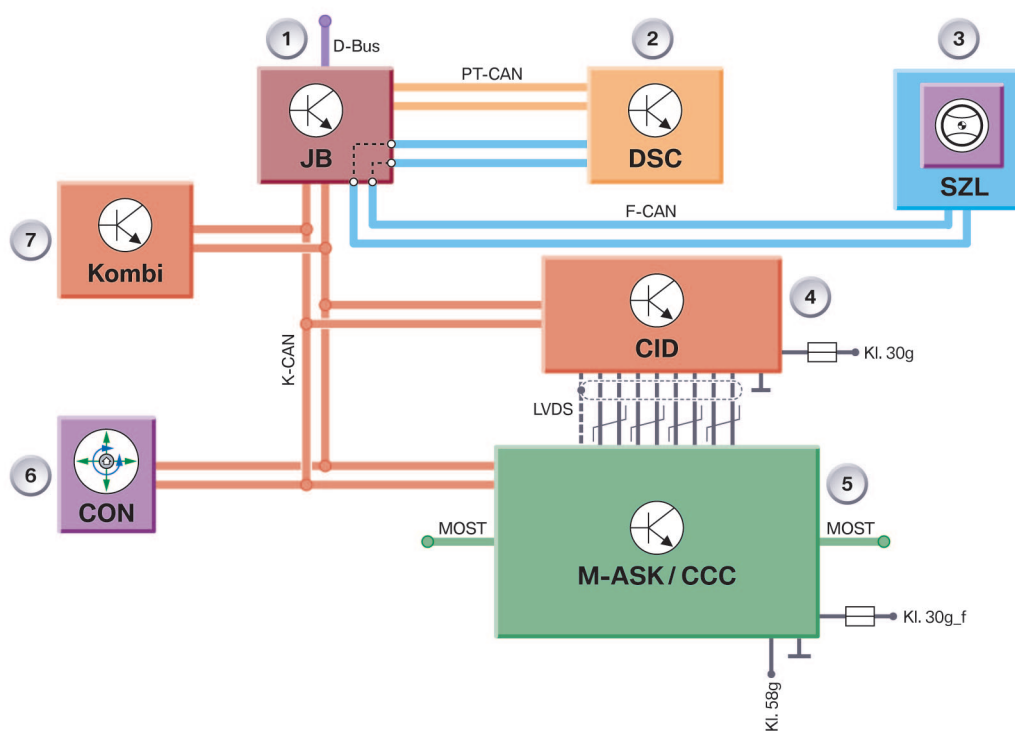


4 - Input/output central information display CID

The input/output key is identical to that of the system circuit diagram and is provided after the system circuit diagram.

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System circuit diagram



TED4-4600

5 - System circuit diagram central information display CID

Index	Description	Index	Description
1	Junction box control unit	5	Multi-audio system controller
2	DSC control unit	6	Controller
3	Multifunction steering wheel	7	Instrument cluster
4	Central information display		



Functions

Displays, Indicators and Controls

Instrument cluster

The instrument cluster receives information on the wiring harness in the form of analogue and digital electrical signals. These signals are processed and displayed in the instrument

cluster or passed on as information to other control units.

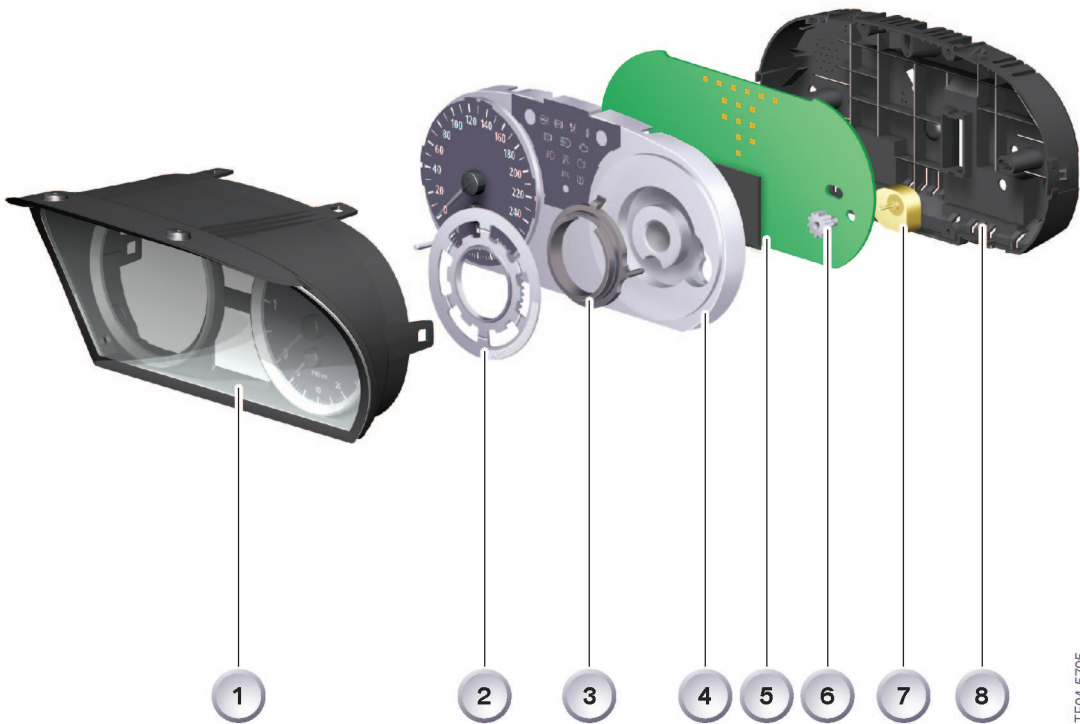
The instrument cluster on the BMW 3 Series features several functions that have been changed compared to previous models.

Moving dial indicator

Movement

Fig. 1 shows the movement sequence of the moving dial indicator based on the rev counter. The moving dial indicator (2) is driven by

- ring gear (3) connected to the dial,
- pinion (6) and
- stepper motor (7) secured to the rear of pc-board (5).



1 - Moving dial indicator in the BMW 3 Series High instrument cluster

Index	Description	Index	Description
1	Front frame	5	PC-board
2	Moving dial indicator	6	Pinion
3	Ring gear	7	Stepper motor
4	Light guide	8	Base plate

Fuel gauge

Fuel reserve



The fuel reserve level is not indicated by an indicator lamp as in the previous models. A fuel pump symbol lights up for 23 seconds in the LC display as soon as the

reserve level is reached.

This display is permanently activated at a range below approx. 50 km.

On-board computer

Main menu

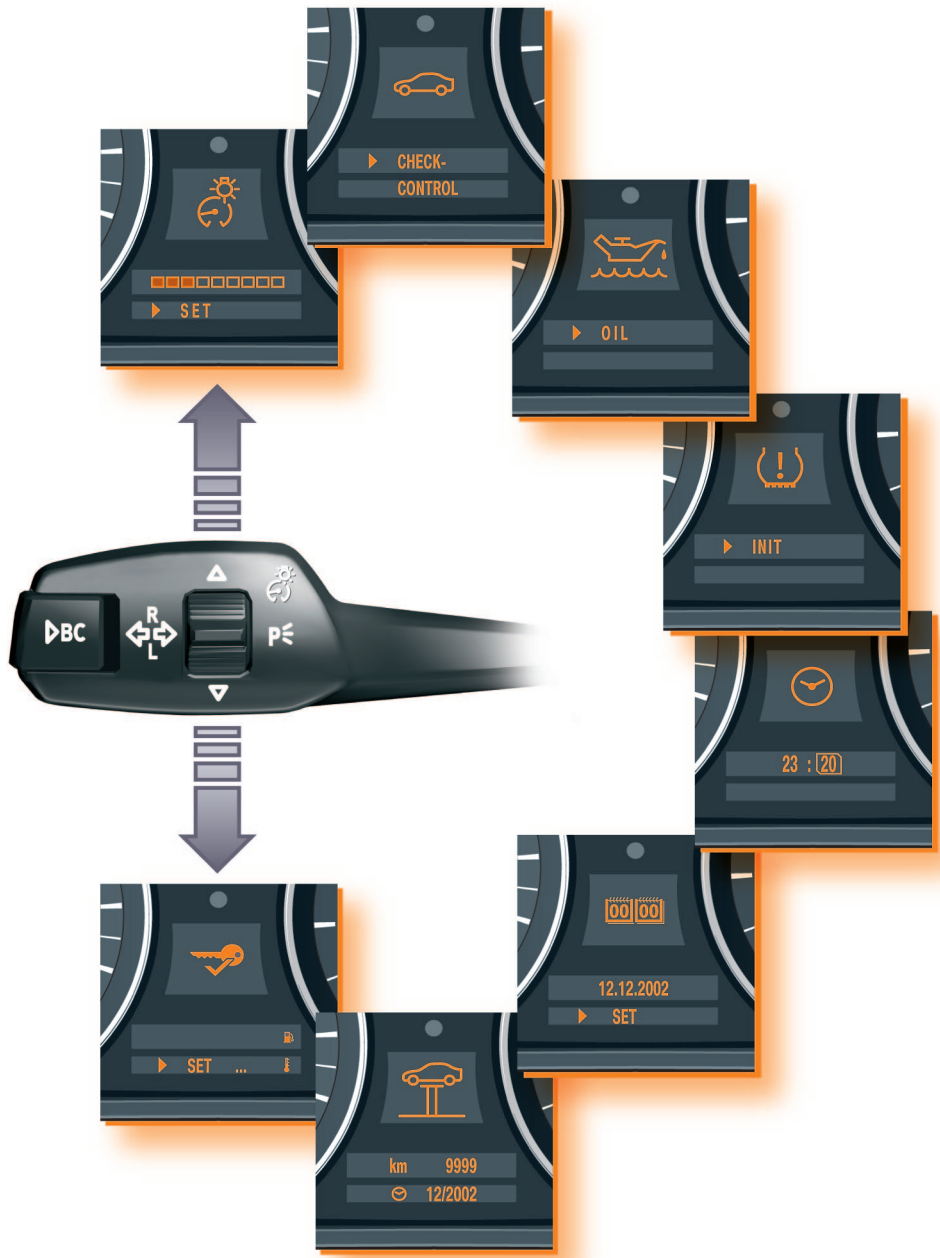
A graphic symbol in the upper display window is assigned to each main menu item. Menu items that are deactivated during vehicle operation are not shown.

Each menu can be interrupted at a certain position by briefly pressing the BC button.

In addition to this active termination, there is an automatic termination that takes place 15 seconds after the last entry.

The display for the M-ASK and CCC are shown on the central information display CID.

BC main menu



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2 - BC main menu

The following table lists all BC functions that can be selected in the instrument cluster depending on the options.

BC function in the instrument cluster

Function	Display	Active as from	Activities	RAD1/2	MASK/CCC
Instrument lighting		BUS active Terminal 58 g	Settings are stored immediately	X	X
Check Control messages		KL. R EIN	Error message system with max. 72 symbols Error prioritization Audible warning Text message as from radio stage 3/4	X	X
Engine oil level measurement		Terminal 15	Measurement with vehicle stationary and in operation Clock symbol in servicing mode No electrical measurement for diesel engines "Service2" displayed in case of fault	X	
Tyre failure indicator RPA		Terminal 15	Multiple initialization possible	X	
Time		KL. R EIN	Menu selection via cursor	X	X
Date		KL. R EIN	Menu selection via cursor	X	X
CBS4 workshop mode		Terminal 15	Activation via reset button ON time > 10 sec	X	
CBS4		Terminal 15	Setting of main inspection/ exhaust emission inspection same as data entry	X	
Personal profile settings		Terminal 15	Changing/resetting units	X	

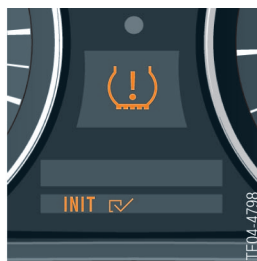
X = Can be selected via instrument cluster

Operating example: Tyre failure indicator RPA

Initialization must be performed immediately after correcting the tyre pressure, a tyre change or a wheel change.

The following procedure must be performed to initialize the system:

1. Start the engine but do not drive off.
2. Press the rocker switch on the steering column stalk until the "initialising tyre failure indicator" function is shown in the instrument cluster.



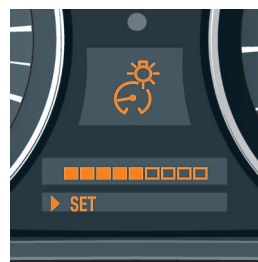
3 - Initialization of the tyre failure indicator RPA

Operating example: Instrument lighting

The side lights or low beam headlights must be switched on in order to control the lighting intensity (dimming).

The following procedure must be performed to set the system:

1. Press the rocker switch in the steering column stalk up or down until the "instrument lighting" function appears in the instrument cluster.
2. Confirm the display by briefly pressing the BC button on the steering column stalk.
3. Press the rocker switch up or down to select the lighting intensity. Each setting is saved immediately.
4. Press the BC button on the steering column stalk to exit the menu.



4 - Setting instrument lighting



The procedure for selecting and correcting the functions

- Check Control messages
- Oil level measurement
- Time
- Date
- CBS4 and
- Personal profile PP settings.

is identical and is described in the BMW 3 Series Owner's Handbook.

Central Information Display (CID)

The central information display CID is designed the same as the CID installed in the BMW 5 Series and BMW 7 Series.

User interface

The user interface in the CID is identical to that of the CID installed in the BMW 5 Series and BMW 1 Series.

A detailed description of the main menu and of the procedure for selecting the individual functions is provided in the BMW 3 Series Owner's Handbook.



5 - User interface in the central information display (CID)

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Personal Profile

Display in the instrument cluster

Display formats and units of measure

The following display formats and units of measure can be changed in the instrument cluster:

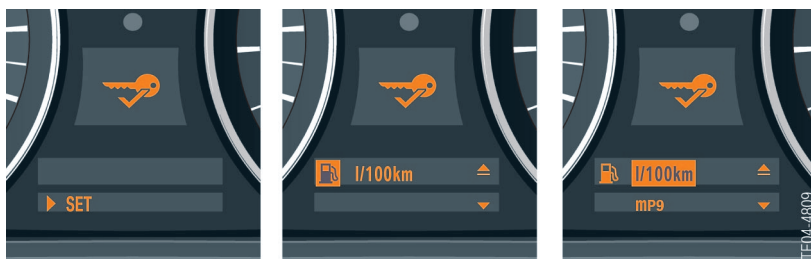
- Fuel consumption (l/100 km, mpg, km/l)
- Distance (km, mls)
- Time format (12h/24h)
- Date format (dd/mm, mm/dd)
- Temperature (°C, °F)
- Reset of display formats and units of measure, the factory setting is adopted.

The procedure for selecting and changing all display formats and units of measure is identical. By way of example, the procedure for changing the unit of measure is described in detail.

Changing unit of measure

The following procedure must be carried out in order to change the unit of measure:

1. Press the rocker switch in the steering column stalk up or down until the "Personal Profile" function appears in the instrument cluster.
2. Press the BC button on the steering column stalk.
3. Press the rocker switch up or down until the required unit of measures is shown.
4. Press the BC button on the steering column stalk.
5. Use the rocker switch to make the necessary change.
6. Press the BC button to adopt the change.



6 - Changing unit of measure



System components

Displays, Indicators and Controls

Great variety of different versions

20 variants of the instrument cluster are available for the E90.

The following table lists the 10 variants that can be installed in the Basic and High versions.

Country-specific variant	Engine variant
E9x ECE LL	4-cylinder petrol engine
E9x ECE RR	4-cylinder petrol engine
E9x US	4-cylinder petrol engine
E9x JPN	4-cylinder petrol engine
E9x ECE LL	6-cylinder petrol engine (NG6)
E9x ECE RR	6-cylinder petrol engine (NG6)
E9x US	6-cylinder petrol engine (NG6)
E9x JPN	6-cylinder petrol engine (NG6)
E9x ECE LL	4/6-cylinder diesel engine
E9x ECE RR	4/6-cylinder diesel engine

The following table provides an exact comparison, illustrating the functions and

symbols that have changed compared to the BMW 3 Series (E46).

Function	E90 Basic	E90 High	E46
			
Analogue instrument	• Speed • Engine speed • Fuel gauge • Economy control	• Speed - moving dial indicator - • Engine speed • Fuel gauge • Economy control	• Speed • Engine speed • Fuel gauge • Economy control • Coolant temperature gauge
Indicator lamps	Maximum 15		Maximum 24
Service indicator	CBS4		SIA4
Check control messages	CC system, multicoloured (yellow/red)		----
Pictographs	Included in CC system		5 door/lid symbols
Door open Light failure			4 lamp failure symbols
Fuel reserve	Contained in CC system (incl. range indication)		Fixed indicator lamp
Unit master	Implemented		Not used
ESS/SSG indicator	Integrated in main display (codeable)		Additional LC display (specific variant)

Function	E90 Basic	E90 High	E46
Main display	Variable indicator lamps (multicoloured) fixed display (2-line) - Basic (0.5 pitch, single colour) - High (0.32 pitch, single colour)		Display with 7 fixed segment LCD (single colour)
BC functions	Standard features - Range - Average speed - Average fuel consumption		Special features (model-specific)
Clock/outside temperature indicator	Permanent in variable indicator field		Integrated in BC
Dimmer (terminal 58g)	BC function Display/indicators in instrument cluster		Thumb wheel in light switch cluster LSZ
Photosensor	Central photosensor in instrument cluster		Central photosensor in light switch cluster LSZ
Gateway function	No (junction box function)		Yes
Oil level measurement	BC function (petrol engine) Indication in instrument cluster		---
Tyre failure indicator RPA	BC function Display/indicators in instrument cluster		Button Centre column switch cluster SZM
Clock	BD function Indication in instrument cluster		Button in instrument cluster
Date	BD function Indication in instrument cluster		Button in instrument cluster
Condition-based service CBS4	BD function Indication in instrument cluster		---
Personal Profile PP	BD function Indication in instrument cluster		Encoding by dealership network (minimum features)

Instrument Cluster

Location/components

Location

The instrument cluster is secured by means of two pan-head tapping screws to the instrument panel.

Components

The instrument cluster comprises the following components:

- Instrument dials
- Indicator and warning lamps
- Program and gear display for automatic transmission and sequential manual gearbox
- Acoustic generator for audible direction indicator signal and CC gong for radio 1 and radio 2.
The audible signals are output via the radio speaker in connection with radio 3 and radio 4.
Driver control request of ACC with radio 1.
- Button for resetting trip distance recorder and selecting condition-based service in CBS menu.
Press button for > 4 seconds to select the workshop menu. The settings are selected via the rocker switch in the steering column stalk.
- Connected components which serve to activate the displays in the instrument

cluster (see system overview/system circuit diagram).

The following components are described in detail:

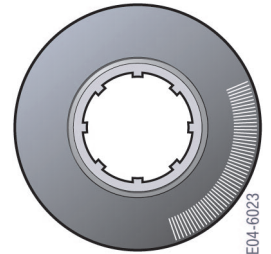
- Display areas
- Indicator and warning lamps
- LC display.

Display areas

The instrument cluster features display areas for:

- Speedometer
- Rev counter
- Economy control (fuel consumption indicator)
- Fuel gauge
- Outside temperature display
- Indicator and warning lamps
- LC display
- Program and gear displays for automatic transmission and sequential manual gearbox SMG.

A shroud prevents reflections in the acutely angled windscreen.



TE04-6023

The set speed for the cruise control and the warning zone of the rev counter are realized by a moving dial indicator in the High variant. The moving dial indicators are located directly behind the speedometer and rev counter scales.



TE04-5792

1 - Pointer instruments, example - High, petrol engine

Speedometer

Two variants of the speedometer are available for the BMW 3 Series. The signal chain is identical to the signal chain of the rev counter.

Variant	Unit	Reading
ECE, JPN	km/h	0·20·40·60·80...260
UK, US	mph	0...20...40...60...160
	km/h	0 20 40 60 80 ... 260

Rev counter

On the BMW 3 Series, engine speed is displayed using the following signal chain:

- The DME control unit sends the engine speed on the PT-CAN and K-CAN.
- Using a characteristic curve, step pulses for actuating the stepper motor are assigned to the effective engine speed.

There are two variants of the rev counter for vehicles with 4-cylinder engines:

- On petrol engines, up to 7000 rpm
- On diesel engines, up to 5500 rpm.

⚠ The engine speed range is increased to 7500 rpm on vehicles equipped with 6-cylinder petrol engines. ◀

Economy control (fuel consumption indicator)

The economy control is an analogue indicator in the instrument cluster.

Fuel gauge

The fuel level is indicated by a pointer instrument integrated in the speedometer on the left.

A pictogram of a fuel gauge lights up in the instrument cluster when the level drops below a factory-coded threshold (standard = 8 l petrol, 6.5 l diesel).

A warning tone additionally sounds on reaching the reserve threshold.

Outside temperature display

A temperature sensor measures the outside temperature and displays it in the instrument cluster.

In ignition key position 0, the instrument cluster applies terminal 30g current to the temperature sensor every 10 minutes.

The instrument cluster makes available the current outside temperature in the form of a data telegram via the K-CAN.

Indicator and warning lamps

The indicator and warning lamps are activated by the processor in the instrument cluster.

All important and legally stipulated indicator and warning lamps are activated at terminal 15 ON during the pre-drive check.



2 - Indicator and warning lamps - Basic variant



3 - Indicator and warning lamps - High variant

The indicator and warning lamps can be illuminated in different colours or combinations.

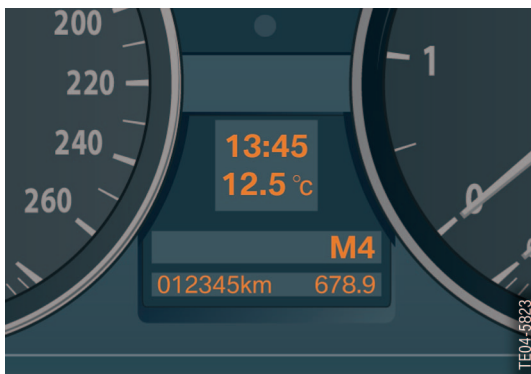
The significance of the indicator and warning lamps as well as the colour assignments are described in detail in the BMW 3 Series Owner's Handbook.

LC display

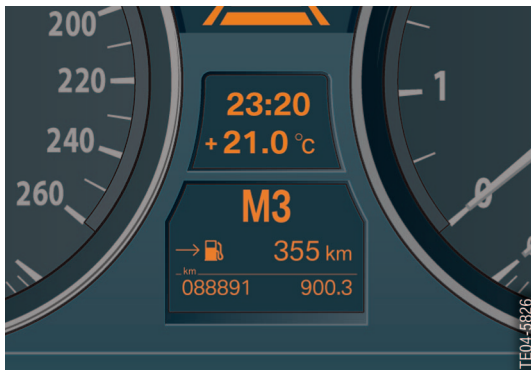
The LC display is divided into two areas.

The time and outside temperature are shown in the upper display along with the CC messages and CBS images.

The on-board computer functions, CBS messages, distance recorder as well as the program display for automatic transmission are shown in the lower display.



4 - LC display - Basic variant



5 - LC display - High variant

Manipulation dot

Different data are stored in the instrument cluster and in the CAS when a dot appears to the left of the trip distance recorder.

The manipulation dot is indicated when, for example, comparison of the stored vehicle identification number does not agree.

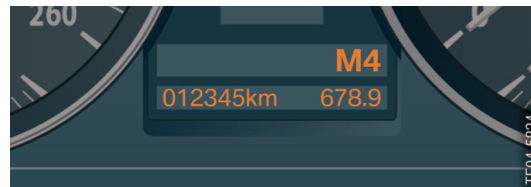
⚠ Different data may be caused, for example, by replacing one of these two control units. ◀

Program and gear display

The program and gear display is shown in the bottom window in the LC display on vehicles with automatic or manual transmission.

The program and gear display shows letters and numbers. The program mode is displayed all the time and is not overwritten by other information.

On manual transmission vehicles, the gear display is blanked out by means of the coding and an enlarged version of the BC display is coded instead of it.



6 - Program and gear indicator - Basic variant



7 - Program and gear indicator - High variant

Acoustic generators

Audible warnings are given in support of check control messages. The instrument cluster controls these warnings via the K-CAN. The warning signals are output by the MASK/CCC control unit when installed as an option.

The acoustic generator in the instrument cluster sounds the warnings if radio 1 or radio 2 is installed in the vehicle.

The footwell module is responsible for control of the direction indicator function via the K-CAN.

Lighting/dimming of display instruments

The instrument cluster is illuminated in the lights ON setting.

The instrument cluster calculates the brightness of the lighting (dimming) of the display, pointer instruments. The brightness setting is distributed over the K-CAN to further control units such as the CID in the vehicle.

When the vehicle lighting is switched on, the dimmer control is matched with the phototransistor integrated in the instrument cluster. The lighting intensity can be set individually in the BC menu via the rocker switch on the steering column.



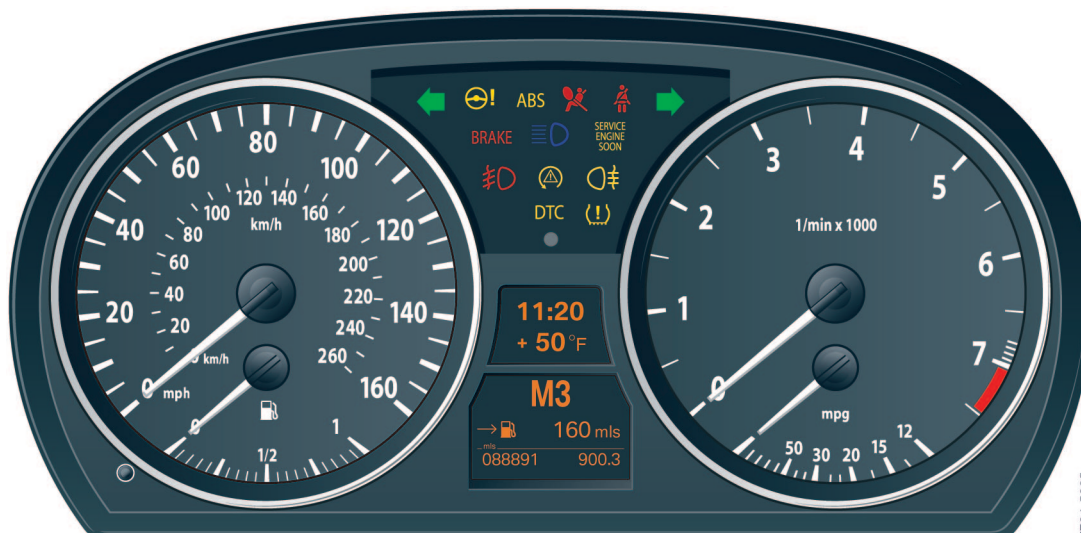
8 - Setting lighting/dimmer via the steering column stalk

When the vehicle lighting is switched off, the system is controlled only by the phototransistor so that the display brightness is determined by the ambient light.

US variant

The main difference compared to the EU LHD instrument cluster is in the speed scale which shows both mph (miles per hour) and km/h

(kilometre per hour). In addition, fuel consumption is shown in mpg (miles per gallon).



9 - Instrument cluster, US variant

There is a further difference compared to the EU variant in the arrangement and representation of the indicator and warning lamps.

The following symbols are replaced by words:

Symbol	Words
	BRAKE
	SERVICE ENGINE SOON

⚠ The indicator lamp for the rear fog light is not active in the US variant. ◀

Audible warnings

US vehicles additionally feature an ignition key warning and a seat belt warning.

An uninterrupted warning tone sounds when the driver's door is opened with terminal 15 OFF and the identification transmitter in place. The audible signal is switched off by removing the identification transmitter, closing the door or after 30 minutes.

The seat belt warning is activated at terminal 15 ON if the seat belt contact is not closed. The audible warning is intermittent and is no longer than 6 seconds. The indicator and warning lamp remains on.

JPN variant

The main difference compared to the EU LHD instrument cluster is in the economy control which shows km/l instead of l/100 km.



10 - Economy control - instrument cluster, JPN variant

On-board computer

There are two versions of the computer available for the BMW 3 Series:

- On-board computer as basic version and on-board computer as journey computer; which can be activated by encoding, depending on the vehicle equipment specification.

The basic on-board computer contains the following functions:

- Average fuel consumption 1
- Range
- Current consumption
- Average speed

For the journey computer, the computer is expanded to include the following additional functions:

- Start of journey
- Duration of journey
- Distance covered
- Arrival time
- Average fuel consumption 2
- Average speed 2
- Remaining distance.

The individual functions of the basic version of the computer can be shown in succession in the LC display of the instrument cluster. The data description is the same as that for the other BMW models.

⚠ The journey computer is shown only in the option SA 555 CID. ◀

Display variants

There are two display variants of the on-board computer functions in the instrument cluster.



1



2

TE04-4872

11 - Display of on-board computer functions

Index	Description
1	Computer in vehicle with automatic transmission
2	Computer in vehicle with manual transmission

The menu rocker switch must be pressed for at least 2 seconds in one direction to enable fast scrolling of all menu items. The menu is then scrolled through at intervals of 0.5 seconds.

Show displays

The BC displays are shown and scrolled in the instrument cluster via a menu rocker switch on the steering column stalk for the direction indicator lights.

The individual functions are displayed in the lower display window of the instrument cluster.

Once terminal R is switched on, the computer will display the computer function that was displayed last.

All other functions can be selected by means of the rocker switch on the steering column stalk.

The order in which the BC functions are shown is always the same.

The main menu is described in detail in the section "Functions".



TE04-4266

12 - Steering column stalk for direction indicator lights

Index	Description
1	BC button
2	Menu rocker switch

Central Information Display (CID)

Location/components

Location

The central information display CID is secured with two pan-head tapping screws in the central area of the instrument panel.



13 - Central Information Display (CID)

Components

The CID comprises the following two components:

- Casing with integral electronic module
- Casing attachment with glass cover.

The controller which is used to control the displays and indicators in the CID is also an integral part of the CID system.

LC display

To accommodate the different equipment variants, two versions of the LC display are available in the new BMW 3 Series:

- LCD 6.5" MID in connection with M-ASK
The display is equipped with a digital 6.5" TFT screen (Thin Film Transistor) with a visible area of 143.4 mm x 79.32 mm. The resolution of the TFT display is 400 x 3 x 240 pixels.
- LCD 8.8" High in connection with CCC
The display is equipped with a digital 8.8" TFT screen (Thin Film Transistor) with a visible area of 209.28 mm x 78.48 mm. The resolution of the TFT display is 400 x 3 x 240 pixels.

Controller

The controller is the central operating control for all comfort functions and selected options for some vehicle functions that are displayed on the central information display.

The controller is located in the centre console immediately behind the gear selector lever, within reach of the user (driver and front passenger).

There are two variants, as follows:

- Basic variant in connection with M-ASK
The basic variant features a notch mechanism with 24 notch settings per revolution.
- High variant in connection with CCC
The tactile feedback for the rotary movement at the controller is generated electrically in the High variant. The tactile feedback for the rest position, the main directions of movement and the depressed position is created by mechanical means.



14 - Controller

Index	Description
1	Controller
2	Button for main menu
3	Button for voice input system

The principle of the controller is identical to that on the BMW 1 Series, BMW 5 Series and BMW 7 Series models.

The menu button which can be used to show the main menu in the CID is additionally located directly behind the controller.

In a second variant, in addition to the menu button, another button is provided to activate/deactivate the voice recognition system SVS.

The button signals are read into the controller and converted into K-CAN telegrams.

ConnectedService

Condition Based Service (CBS)

In the same way as the mid-range and luxury models, the BMW 3 Series will also be offering service intervals that are geared towards the current condition of selected components, i.e. servicing that depends on the condition of components and maintenance requirements; this is known as Condition Based Service (CBS).

Condition based service therefore means - servicing only when a part is worn.

The system calculates when a service operation, e.g. an oil change is due and informs the customer of this via the LC display in the instrument cluster.

CBS sorts all the data according to date due and can determine when the vehicle should be taken into service.

If a central information display CID is installed, the instrument cluster passes on the sorted data to the CID where the data can be displayed in the service menu.

There are ten different types of service, with each of these types being assigned to a specific service group, e.g. oil change.

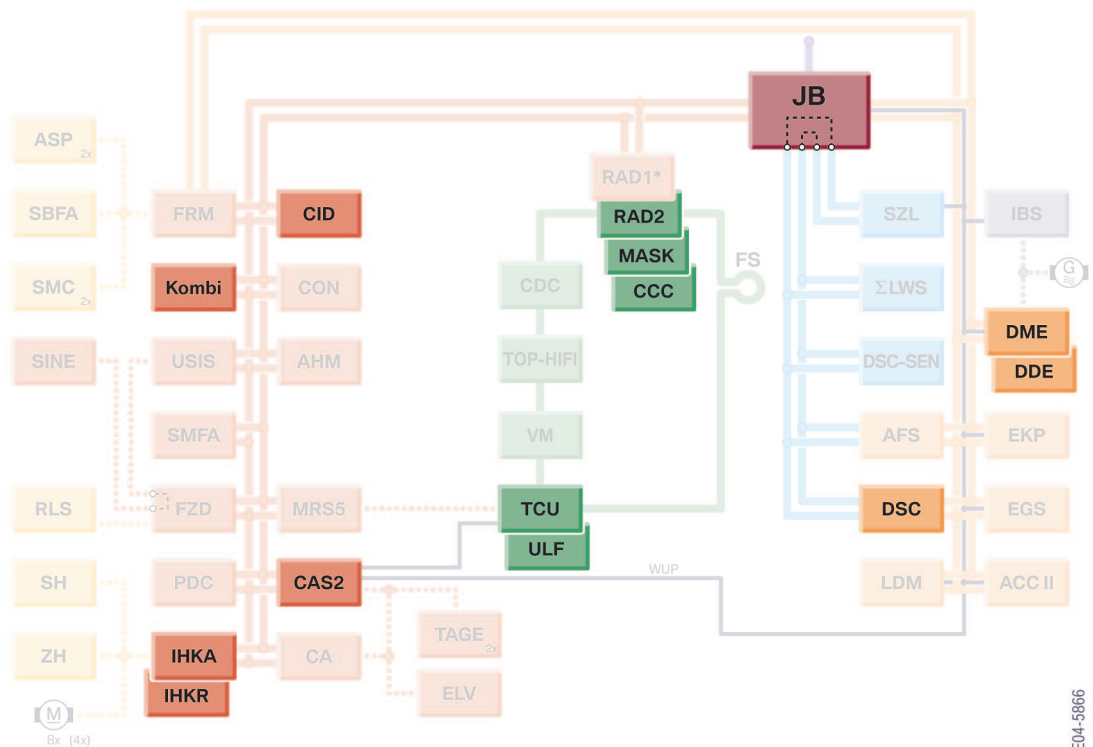
The system comprises the following components with their sensors:

- Instrument cluster
- External control units such as DME, IHKA and DSC
- Car Access System (CAS) 2
- Central information display CID.

Block diagram EU

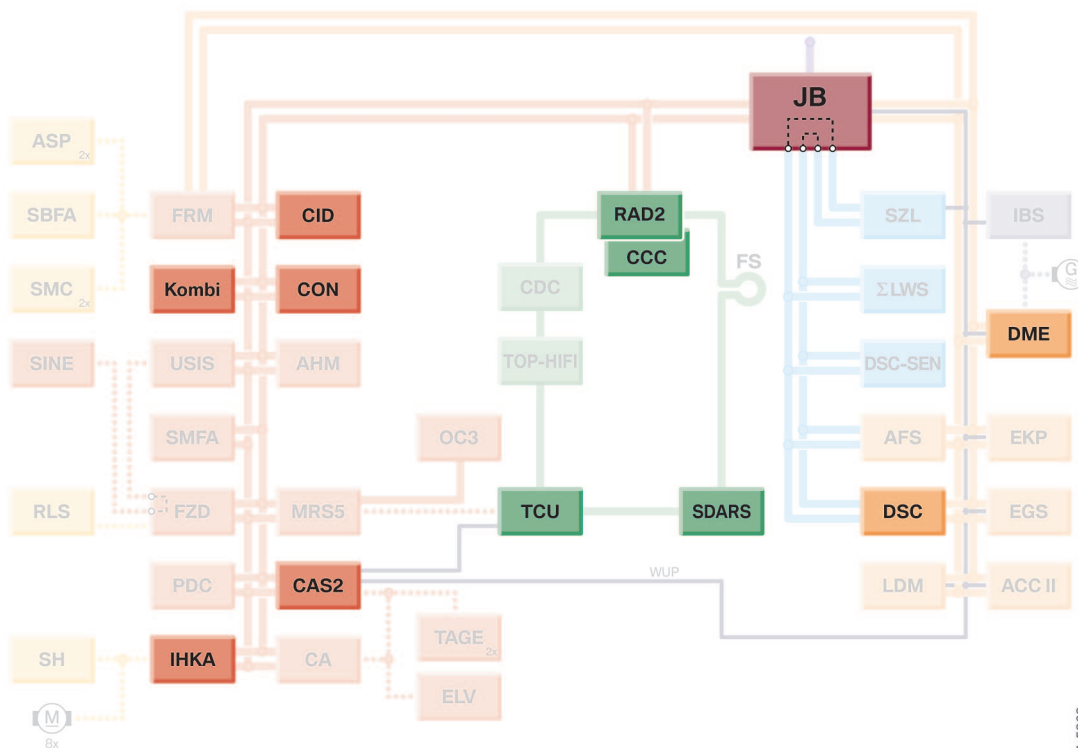
All information that the CBS system requires is sent on the K-CAN bus .

The CBS requests from all the control units are therefore sent via the K-CAN to the instrument cluster and to the Central Information Display.



15 - Block diagram of the Condition Based Service CBS EU

Block diagram US



TE04-5868

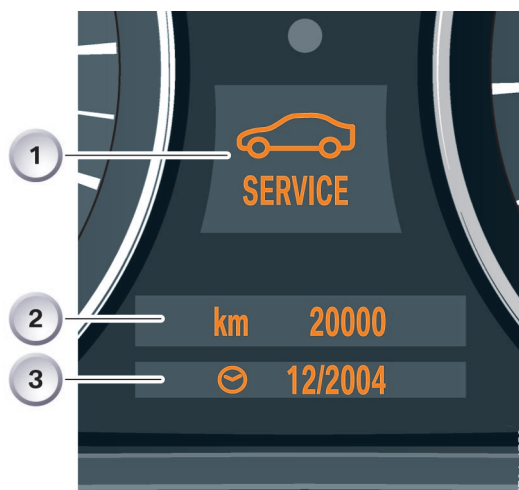
16 - Block diagram of the Condition Based Service CBS US

CBS displays in the instrument cluster

The CBS display always comprises the following two separate displays:

- A coloured symbol in the upper display
 - Orange for normal
 - Yellow for service due
 - Red for service overdue
- and information on remaining distance and/or due date in the lower display.

You will find further information on the CBS displays in the BMW 3 Series Owner's Handbook under Servicing Systems.

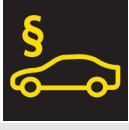


TE04-4763

17 - CBS display in the instrument cluster

Index	Description
1	CBS symbol
2	Remaining distance display
3	Final date information

CBS symbols

Normal condition		Service due		Service overdue	Explanation
					Engine oil
					Micro-filters
					Front brakes
					Rear brakes
					Brake fluid
					Vehicle check
					Spark plugs
					General inspection
					Exhaust-gas test
					Particle filter

Displays in the Central Information Display

All information on the individual service operations can be displayed in the CID.

The CBS functions are accessible under the <Settings> menu item.

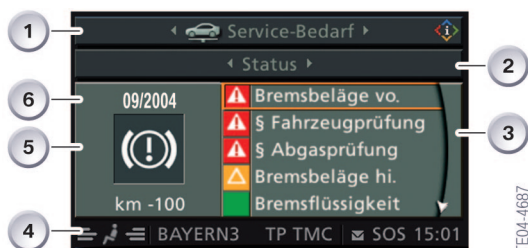
Selecting CBS menu

Press the controller, the "Settings" menu will appear.

Turn the controller until "Service" is highlighted, then press to activate the CBS menu.

The following operating and display fields are shown:

- Status bar
- Menu bar 1
- Menu bar 2
- Display field for CBS symbol
- Display field for service operation.



18 - Service menu

Index Description

1	Menu bar 1 - Service requirement, CC messages - Service
2	Menu bar 2 - Status - There are currently no requirement messages
3	Display field for CBS symbols
4	Status bar
5	Display field for service operation
6	Target date

The service operation display field always shows the first five messages.

Any overdue service operations and symbols marked in red in the list are always at the top of the list of messages.

To display the information concerning a specific service operation on the CID, turn the controller to select the required service operation and display the selection by pressing the controller.

The following display appears in the CID if, for example, the service operation "§ Vehicle inspection" is selected and activated:



19 - Example "Legally required vehicle inspection"

Index Description

1	Exit display, return to last setting
2	Date for general inspection
3	Text field for further information
4	Schedule date for service

Once the service date has been set and confirmed by pressing the controller, the priority of the service operation changes, e.g. from red to green and the service operation is sorted in the list.



Service information

Displays, Indicators and Controls

Instrument Cluster

Information for technicians

Test functions

The test functions are shown in the LC display of the instrument cluster.

The test functions are used by the BMW service mechanics to check the coding. They also provide help in troubleshooting without the diagnostic tester.

To start function test

- Terminal R ON or terminal 15 ON
- Press and hold the reset button in the instrument cluster for 10 seconds (set/reset).

⚠ The test functions can also be called up by holding down the setting button in the instrument cluster and simultaneously switching on terminal R. ◀

Display of test functions

The test functions are shown only in the upper LC display.

Locking and unlocking the test functions (test function 19)

Only the first two test functions are freely accessible.

As from the third, all further test functions are locked. The functions can be unlocked only via test function 19.

The test functions are unlocked by entering the sum of the digits in the vehicle identification number.

To end test function

- Ignition key at terminal R or terminal 15 ON
- Press and hold the setting button for longer than 5 seconds or
- calling up test function 21 (RESET).

⚠ To protect against unauthorized access, all test functions (with the exception of test 1 and test 2), are locked again after a RESET. ◀

Visual system test

In the visual system test, all the indicator lamps and lights are lit briefly.

The needle instruments are moved from the lower to upper stop and back again.



All the described test functions can also be performed via the BMW diagnostics system.



TE04-5867

1 - Test function 02, visual system test

Overview of test functions

Only the main test functions are listed in the following table. In addition to the majority of test functions there are further equivalent

functions for which a similar display appears in the instrument cluster.

Test function	Description	Display
01	Instrument cluster identification - Vehicle identification number, last 5 digits	01.00 FGSTNR AB12345
02	System test	02.00 KI TEST
03	Not used	03.00 Not used
04	Electric load values	04.00 VERB-MOM 12,6 l/100 km
05	Range consumption	05.00 RW-VERBR
06	Fuel level	06.00 TANK L R S 24.5 26.7 50
07	Current display values	07.00 KTMP-MOM 104°C
08	Speed	08.00 V-EFF 123 km/h
09	System voltage	09.00 Ub 13.3 V
10	Not used	10.00 Not used
11	Units	11.00 ZEIT-EINH 24h
12	Calculated time of arrival	12.00 V-ANKUNFT 67,8 km/h
13	Audible signals	13.00 AUDIO LICHT-WARN BLINKER ZS-WARN
14	Self-diagnostics	14.00 FSP-Einträge 10
15	I/O ports processor	15.00 PORT 00 01010111
16	Dimming	16.00 DIMMRAD-CAN 46h

Test function	Description	Display
17	Contrast	17.00 DISP-HEIZ Ein io
18	Not used	18.00 Not used
19	Locking	19.00 LOCK LOCK: ON LOCK. 25
20	Fuel consumption correction	20.00 KORR-VERBR 1000 1er KORR 10er KORR 100er KORR
21	Reset (software reset)	21.00 Reset?

Component replacement and trial replacement

There are three possible combinations for replacing the instrument clusters and car access system 2 CAS 2.

- Instrument cluster defective, CAS 2 OK
- CAS 2 defective, instrument cluster OK
- CAS 2 and instrument cluster must be replaced.

Simultaneous replacement of CAS 2 and the instrument cluster should be avoided. The odometer reading will be lost as a result.

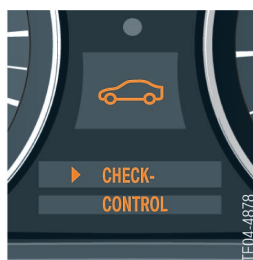
In principle, it is also possible to carry out a trial replacement of the instrument cluster and CAS 2.

To select check control messages in the instrument cluster

The instrument cluster shows a series of vehicle statuses and consequently generates CC messages.

The following procedure must be performed to select a CC message:

1. Press the rocker switch in the steering column stalk up or down until the "Check Control" function appears in the instrument cluster.
2. Press the BC button on the steering column stalk and confirm the display.
3. Press the rocker switch up or down to select the CC messages.
4. Press the BC button on the steering column stalk to exit the menu.



2 - CC Message

⚠ The outside temperature or a previous priority 1 message is displayed again if the rocker switch is not operated for 8 seconds.



Central Information Display (CID)

Information for technicians

Service mode

The controller can be used to activate the Service mode functions.

Service mode is a special facility which provides information about the status of the display and user control system.

Service mode can be used, for example, to read out hardware/software statuses for the central information display or control units in the M-ASK/CCC system network.

As an addition to the comprehensive facilities of the diagnosis system, Service mode acts as a simple means of quickly accessing diagnostic data without the need for a diagnosis tester.



3 - Service mode

Sources of information

In Service mode, information on the following functions can be selected from the upper menu bar:

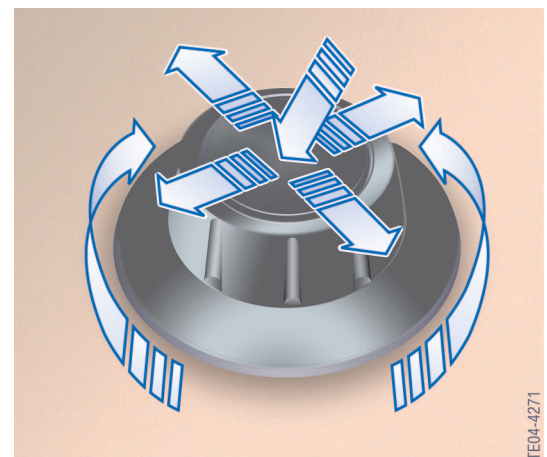
- Radio
- Version control
- Navigation
- GPS
- Sensor test.

Push the controller in any direction to return to the main menu.

Activating Service mode

Select main menu and press and hold the controller. Tactile feedback will then be generated.

- Turn controller 3 stops clockwise
- Turn controller 3 stops anticlockwise
- Turn controller 1 stop clockwise
- Turn controller 1 stop anticlockwise
- Turn controller 1 stop clockwise
- Press the controller to confirm, Service mode will then appear in the CID.



4 - Activating Service mode

ConnectedService

Information for technicians

Resetting the service operations

When one or more service operations have been carried out, e.g. front brake pads have been changed, these operations must be reset to their full service interval.

There are two options for resetting the service operations:

1. Legally required service operations such as the vehicle inspection (HU) and exhaust emission inspection (AU) can only be reset in the "Service" menu.
2. All vehicle servicing service functions such as changing spark plugs are reset via the reset button for the trip distance recorder in the instrument cluster. If the reset button is pressed for longer than ten seconds, the reset mode opens automatically.

"Reset?" is displayed in the lower display window.

In the upper display window, the CBS symbol, e.g. for "engine oil service overdue" will be displayed.

Press the reset button until the time/distance-dependent displays in the lower display window are replaced with dashes.

Reset is no longer possible once 80 percent of the interval has expired. A reset lock will be shown in the display with "OK".

Entering due date

The due date for the legally required general inspection and exhaust emission inspection can be entered only in the central information display with the aid of the controller.

Since different laws are applicable depending on the country, country-specific intervals can be found at this point. For markets where no such regulation applies for general inspection and exhaust test, this function can be suppressed by the software.

Carry out the following procedure to enter the due date:

- Select "Service" from the "Settings" menu and confirm
- Select service operation "§ Vehicle inspection" for example and confirm. "Set service date" is marked.
- Press controller to activate the input box.
- Enter the date by turning and pressing the controller.
- Select "Exit display" and confirm to return to the last setting.



5 - Entering due date

Index	Description
1	Exit display, return to last setting
2	Date for statutory vehicle inspection
3	Text field for further information
4	Activate deadline in (2)



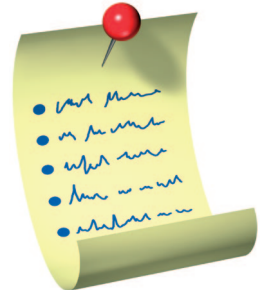
Summary

Displays, Indicators and Controls

Points to remember

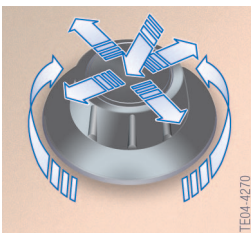
The most important information concerning the display, indicator and control concept in the BMW 3 Series is summarized in the following table.

The list is intended to provide you with the contents and important points of this Participant's Manual in concise form.



Theoretical and practical notes for day to day use.

Operating concept

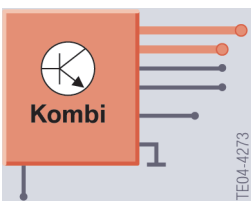


It is possible to operate all comfort functions and several vehicle functions via the controller in the BMW 3 Series.

Instrument cluster

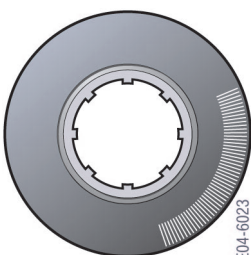


A classic two-dial design draws attention to both instruments. Compact display and indicator units convey the most important information to the driver at a glance.



The instrument cluster has no gateway function as on the previous model series.

Only the outside temperature, the information from the steering column switch cluster and footwell module are shown directly in the display.



In the High variant, the set speed for the cruise control is realized by a moving dial indicator.

The moving dial indicators are located directly behind the speedometer and rev counter scales.



The analogue indicator for the fuel reserve is located in the instrument cluster.

The fuel reserve level is not indicated by an indicator lamp as in the previous models.

A fuel pump symbol is illuminated for 23 seconds in the LC display as soon as the reserve level of 8 or 6.5 litres respectively is reached. This display is permanently activated at a range below approx. 50 km.

Operating concept - steering column stalk



A graphic symbol in the upper display window is assigned to each main menu item.

Menu items that are deactivated during vehicle operation are not shown.

Each menu can be interrupted at a certain position by briefly pressing the BC button.

Depending on the equipment variant and on the radio/navigation systems, several menu items are blanked out and can therefore only be adjusted via an external control unit. The display for the M-ASK and CCC are shown on the central information display CID.

Central information display



The new BMW 3 Series offers the same convenient operating and control functions as the mid-range and luxury BMW models.

The central information display CID is designed the same as the CID installed in the BMW 5 Series and BMW 7 Series.

The user interface in the CID is identical to that of the CID installed in the BMW 5 Series.

Personal Profile



Through the Personal Profile, the BMW 3 Series also caters for the driver's personal requirements.

The Personal Profile allows the driver to set several functions of the BMW 3 Series to suit his/her personal requirements.

Personal Profile stores the data entered by the driver such as automatic setting of the outside mirrors or speed-dependent volume in the corresponding control units.



All test functions described in the Participant's Manual can also be performed via the BMW diagnostics system.



ConnectedService is made up of several modules; some of these are already in operation in other BMW models while others have been further developed or introduced for the first time with the launch of the BMW 3 Series.



It is important that you work through the training and information program (SIP) on this subject. Fundamental knowledge ensures competence in practical applications.



Test questions

Displays, Indicators and Controls

Questions

The knowledge gained can be examined in this section.
Questions on the topic of display, indicator

and control elements are provided in the following for the purpose of consolidating the subject matter.



Extend and consolidate the knowledge gained.

Instrument cluster

Question 1

How many analogue instruments are there in the instrument cluster in the new BMW 3 Series?

Question 2

What type of service indicator is used on the new BMW 3 Series?

Question 3

How can the lighting intensity level of the controls be individually adapted?

Question 4

What information is directly evaluated and displayed in the instrument cluster?

Question 5

What control unit is responsible for the gateway function in the new BMW 3 Series?

Question 6

What functions are indicated by a moving dial indicator in the High variant?

Question 7

How is the fuel reserve shown in the new BMW 3 Series?

Question 8

Explain the signal chain of the rev counter.

Question 9

What symbols are replaced by words in the US variant?

Question 10

What audible warnings additionally sound in the US variant?

Central Information Display (CID)

Question 1

Name the two sizes of LC display that can be installed in the central information display CID of the BMW 3 Series?

Question 2

What control units and controls are connected directly to the central information display CID?

Question 3

What procedure is necessary to call up service mode via the central information display CID?

Question 4

How many menu items does the main menu comprise in the central information display?

ConnectedService

Question 1

What does ConnectedService mean?

Question 2

How many components and their sensors are involved in the BMW maintenance system - condition-based service CBS4?

Question 3

Under what menu item can CBS functions be found?

Question 4

Where can the legally required service functions be reset?

Answers to questions

You will find the answers to the questions on "Displays, indicators and controls" in this section

Instrument cluster

Question 1

The instrument cluster has four analogue instruments.

Question 2

CBS 4 instead of SIA 4

Question 3

By means of the steering column stalk.

Question 4

The outside temperature, the information of the steering column switch cluster SZL and of the footwell module FM are displayed directly in the instrument cluster.

Question 5

The junction box JB is responsible for the gateway function in the new BMW 3 Series.

Question 6

The set speed for the cruise control and the warning zone of the rev counter (09.05) are realized by a moving dial indicator in the High variant.

Question 7

The fuel reserve is an on-board computer function in the new BMW 3 Series.

Question 8

The DME control unit sends the engine speed on the PT-CAN and K-CAN.

Question 9

BRAKE and SERVICE ENGINE SOON.

Question 10

US vehicles additionally feature an ignition key warning and a seat belt warning.

Central Information Display (CID)

Question 1

LCD 6.5" and 8.8".

Question 2

Instrument cluster, JB, DSC, SZL, CON, M-ASK/CCC.

Question 3

Turn controller 3 clicks to the right, 3 to the left, 1 to the right, 1 to the left and 1 to the right. Press the controller to confirm setting.

Question 4

The main menu in the central information display CID comprises 4 menu items.

ConnectedService

Question 1

ConnectedService means "connected service".

Question 2

- Instrument cluster
- External units such as DME, IHKA
- Car access system CAS 2

Central information display CID.

Question 3

The menu item can be found under "Service" in the "Settings" menu.

Question 4

All legally required service functions can be reset only in the "Service" menu.

